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PROJECT REPORT

STREAMLINING IT PROCUREMENT: AUTOMATING STANDARD LAPTOP ORDERS WITH FLOW DESIGNER

1. INTRODUCTION

1.1 Project Overview

IT departments regularly receive requests from employees for laptops and other IT equipment. When these requests are handled manually, the process may involve multiple emails, approvals, procurement activities, and status updates.

The project **“Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer”** focuses on automating the standard laptop procurement process using **ServiceNow Flow Designer**.

The automated workflow manages the process from laptop request submission to approval, procurement task creation, notification, order tracking, and completion.

1.2 Purpose

The purpose of the project is to create a simple and automated workflow for standard laptop procurement.

The system aims to:

- Automate laptop request processing.
- Reduce repetitive manual work.
- Automate approval activities.
- Create procurement tasks automatically.
- Send automatic notifications.
- Track laptop orders.
- Improve visibility of request status.

2. IDEATION PHASE

2.1 Problem Statement

The traditional laptop procurement process often requires employees to submit requests manually and IT teams to perform several activities such as checking request details, obtaining approval, creating procurement tasks, and updating employees.

Manual processing can result in:

- Delayed approvals.
- Repeated data entry.
- Difficulty tracking requests.
- Delayed communication.
- Increased administrative effort.

Therefore, an automated workflow is required to streamline the standard laptop procurement process.

2.2 Empathy Map Canvas

The empathy map identifies the needs and concerns of people involved in the procurement process.

Category	Observation
Says	“I need a laptop for my work.”
Thinks	“How long will it take to receive my laptop?”
Does	Submits a laptop request and checks its status.
Feels	Concerned about delays and lack of status updates.
Pain Points	Manual requests, delayed approval, lack of tracking.
Needs	Quick request processing, approval, notifications, and order tracking.

2.3 Brainstorming

During brainstorming, several ideas were identified to improve IT procurement:

1. Online laptop request form.
2. Automated request validation.
3. Standard laptop selection.
4. Automated manager approval.
5. Automatic procurement task creation.
6. Automatic email notifications.
7. Laptop order status tracking.
8. Procurement dashboard.
9. Automated completion notification.

The selected solution combines these ideas using **ServiceNow Flow Designer**.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	Employee Action	System Action	User Experience
Request	Submits laptop request	Creates request	Easy
Validation	Provides required details	Validates information	Simple
Approval	Waits for approval	Sends request to manager	Automated
Procurement	Waits for laptop	Creates procurement task	Trackable
Order	Checks status	Updates order status	Transparent
Delivery	Receives laptop	Marks request completed	Satisfactory

3.2 Solution Requirement

Functional Requirements

- Employee should be able to submit a laptop request.
- System should validate request information.
- System should check standard laptop availability.
- System should send the request for approval.
- Manager should be able to approve or reject the request.
- System should create a procurement task after approval.
- System should send automatic notifications.
- IT team should be able to update order status.
- System should track the request until completion.

Non-Functional Requirements

- The workflow should be easy to use.
- The system should provide reliable automation.
- Requests should be processed consistently.

- Information should be accessible to authorized users.
- The workflow should be maintainable and scalable.

3.3 Data Flow Diagram

Level 0 Data Flow

Employee

↓

Laptop Request

↓

ServiceNow Procurement System

↓

Flow Designer

↓

Manager Approval

↓

IT Procurement Team

↓

Laptop Order

↓

Employee

Main Data Flow

Employee → Submit Request → Validate Request → Approval → Procurement Task → Order Processing → Delivery → Completion

3.4 Technology Stack

Component	Technology
Platform	ServiceNow
Automation	Flow Designer
Workflow	ServiceNow Flow
Request Management	ServiceNow
Approval	Flow Designer Approval
Notification	ServiceNow Notification
Data Storage	ServiceNow Tables
Testing	ServiceNow Flow Designer

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem	Proposed Solution
Manual laptop requests	Online request form
Manual validation	Automated validation
Delayed approvals	Automated approval flow
Manual task creation	Automatic procurement task
Lack of communication	Automatic notifications
Difficult tracking	Request and order status tracking
Repetitive IT activities	Flow Designer automation

The proposed solution directly addresses the major problems in the existing manual procurement process.

4.2 Proposed Solution

The proposed system uses **ServiceNow Flow Designer** to automate the standard laptop procurement process.

The workflow starts when an employee submits a laptop request.

Workflow

Submit Laptop Request

↓

Validate Request

↓

Check Laptop Availability

↓

Send for Manager Approval

↓

Approval Decision

↓

Create Procurement Task

↓

Send Notification

↓

Process Laptop Order

↓

Update Order Status

↓

Complete Request

4.3 Solution Architecture

The solution consists of the following components:

1. Employee/User

The employee submits a standard laptop request.

2. ServiceNow Request

The request stores employee and laptop information.

3. Flow Designer

Flow Designer automatically controls the procurement workflow.

4. Approval Module

The request is sent to the appropriate manager for approval.

5. Procurement Task

After approval, a procurement task is automatically created.

6. Notification System

Notifications are sent to the employee regarding request status.

7. Order Tracking

The IT team updates and tracks the laptop order until completion.

Architecture Flow

Employee → ServiceNow Request → Flow Designer → Approval → Procurement → Notification → Order Tracking → Completion

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project follows the **Agile Scrum methodology**.

The work is divided into four sprints.

Sprint	Main Activities	Story Points
Sprint-1	Laptop Request and Validation	7
Sprint-2	Flow Automation and Approval	11
Sprint-3	Order Tracking and Testing	8
Sprint-4	Dashboard and Final Validation	8
Total		**34**

Average Velocity

Average Velocity = Total Story Points / Number of Sprints

= 34 / 4

= 8.5 Story Points per Sprint

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Performance testing is performed to verify that the automated workflow functions correctly and processes requests efficiently.

Test Case	Expected Result	Status
Submit laptop request	Request should be created	Pass
Validate request	Request should be validated	Pass
Check availability	Laptop availability should be verified	Pass
Approve request	Workflow should continue	Pass
Reject request	Request should be marked rejected	Pass
Create procurement task	Task should be created automatically	Pass
Send notification	Employee should receive notification	Pass
Update order status	Status should be updated	Pass
Complete request	Request should be marked completed	Pass

7. RESULTS

7.1 Output Screenshots

The following screenshots can be included in the project report:

1. **ServiceNow Login/Home Page**
2. **Laptop Request Form**
3. **Submitted Laptop Request**
4. **Flow Designer Workflow**
5. **Approval Request**

6. **Procurement Task**
7. **Notification**
8. **Laptop Order Status**
9. **Completed Request**
10. **Procurement Dashboard**

Result

The project successfully demonstrates an automated standard laptop procurement workflow using ServiceNow Flow Designer.

The workflow connects request submission, validation, approval, procurement task creation, notification, and order tracking into a single automated process.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Reduces manual work.
- Automates repetitive procurement activities.
- Provides faster request processing.
- Automates approval handling.
- Creates procurement tasks automatically.
- Provides automatic notifications.
- Improves request tracking.
- Provides better process visibility.
- Standardizes laptop procurement.

Disadvantages

- Requires ServiceNow platform access.
- Initial workflow configuration requires technical knowledge.
- Changes in procurement policies may require workflow updates.
- Automation depends on correctly configured ServiceNow data and rules.

9. CONCLUSION

The project **“Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer”** demonstrates how ServiceNow Flow Designer can be used to automate the IT laptop procurement process.

The workflow manages the complete process from request submission to approval, procurement task creation, notification, order tracking, and completion.

The solution reduces repetitive manual activities and provides a structured and trackable procurement process.

10. FUTURE SCOPE

The project can be enhanced in the future by adding:

- Automated inventory management.
- Integration with laptop vendors.
- Automatic purchase order generation.
- Real-time inventory availability.
- Mobile notifications.
- Advanced procurement dashboards.
- AI-based laptop recommendations.
- Automatic delivery tracking.
- Integration with asset management.
- Analytics for procurement performance.

11. APPENDIX

Source Code / Flow Configuration

The project does not require traditional programming code. The main implementation is performed using **ServiceNow Flow Designer**, including triggers, actions, conditions, approvals, tasks, and notifications.

Dataset Link

Not Applicable – This project uses ServiceNow records and workflow data rather than a traditional machine-learning dataset.

GitHub Link

https://github.com/monisha190707-art/Streamlining-IT-Procurement-Automating-Standard-Laptop-Orders-With-Flow-Designer

Project Demo Link

https://drive.google.com/drive/folders/1b-RGfNX9czCB5CTZOnlJEXrogpAjt7JO?usp=drive_link